

Notation Reference

This paper intentionally minimizes mathematical notation to maximize accessibility. Where notation is used, it follows the Cognitive Integrity Framework (CIF) formal specification defined in Part 1 of this series.

Minimal Notation Used

Minimal Notation Used I

Symbol	Meaning	Plain Language
α	Trust decay factor	"Delegated trust decreases by this factor at each step"
n	Agent count	"Number of agents in the system"
f	Byzantine agents	"Maximum number of malicious agents tolerated"
$[A]$	Design Matrix	Maps Functional Requirements to Defense Provisions
$\{FR\}$	Functional Requirements	What the system must protect
$\{DP\}$	Defense Provisions	What CIF mechanisms provide
INV_k	Individual invariant predicate	"A hard rule the system checks at runtime; Part 1 writes this I_k "
Ω_k	Adversary class k	Capability tier: 1=external (input control), 2=peripheral (tool/data channels), 3=agent-level (single compromised agent), 4=coordination (inter-agent channels), 5=systemic (orchestrator)

CIF-AD-OODA Notation

The cross-domain analysis (Sections 9c–9l) uses the CIF-AD-OODA methodology:

- ▶ **Design Matrix** $[A]$: A matrix where rows represent Functional Requirements (FR) and columns represent Defense Provisions (DP). Each entry A_{ij} indicates whether defense j covers requirement i .
- ▶ **Transient Coupling** $[A']$: The coupling matrix during an active attack, showing which defenses are bypassed.
- ▶ **Adversary Classes** Ω_1 – Ω_5 : Five adversary classes from external input control (Ω_1) through peripheral tool/data-channel compromise (Ω_2), single compromised agents (Ω_3), coordination-channel attacks (Ω_4), and systemic orchestrator compromise (Ω_5), as defined in Part 1's threat model.

Three Universal Attack Patterns I

Across all ten domains, attacks reduce to three canonical patterns:

Three Universal Attack Patterns I

Pattern	Description	Defense
FR Polarity Inversion	Attacker flips a Functional Requirement's sign (e.g., "don't share secrets" → "share secrets")	Cognitive Firewall + Belief Sandbox
Constraint Relaxation	Attacker weakens a safety constraint's boundary	Invariant Monitor + Tripwire
Context Boundary Violation	Attacker exploits scope leakage between agent contexts	Provenance Tracking + Trust Calculus

Trust Decay Explanation

Trust Decay Explanation I

The symbol δ is a parameter, not a universal constant. For an illustrative example, $\delta = 0.8$ means:

Trust Decay Explanation I

- ▶ Direct trust: 100% of assigned value
- ▶ One delegation: 80% of source trust
- ▶ Two delegations: 64% of source trust
- ▶ Three delegations: 51.2% of source trust

Trust Decay Explanation I

The executable Part 3 deployment profiles use $\delta = 0.80$ for balanced operation and $\delta = 0.60$ for high assurance. Those are implementation defaults, not a claim that either value is optimal for every threat model. A lower δ means faster decay, providing more security against long delegation chains while limiting delegation utility.

Byzantine Tolerance Explanation

Byzantine Tolerance Explanation I

When we say $n \geq 3f + 1$:

Byzantine Tolerance Explanation I

- ▶ To tolerate 1 malicious agent, need at least 4 agents
- ▶ To tolerate 2 malicious agents, need at least 7 agents
- ▶ To tolerate 3 malicious agents, need at least 10 agents

Full Notation Reference

For complete formal definitions of all CIF notation, see Part 1 supplementary **S03** (`cogsec_multiagent_1_theory/manuscript/S03_notation.md` in this repository's `cognitive_integrity` program tree).

The Part 1 specification uses on the order of 100 symbols covering:

Full Notation Reference I

- ▶ Agent cognitive state
- ▶ Trust calculus operations
- ▶ Defense mechanism parameters
- ▶ Consensus and coordination
- ▶ Information-theoretic bounds